

Quantum Circuit Design using Complex-Valued Neural Network in Stiefel Manifold



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1. Introduction

Quantum computing uses quantum gates to perform operations on qubits via unitary transformations [1]. Designing efficient, hardware-implementable quantum circuits from algorithm descriptions remains a challenge.

- **Goal:** Automate quantum circuit synthesis using machine learning.
- **Challenge:** Maintain unitarity during learning—a fundamental physical constraint [1].
- **Why ML?** Neural networks can learn complex unitary mappings from input–output quantum state pairs [2].
- **Innovation:** We propose a complex-valued neural network trained on the Stiefel manifold to preserve unitarity at every step.

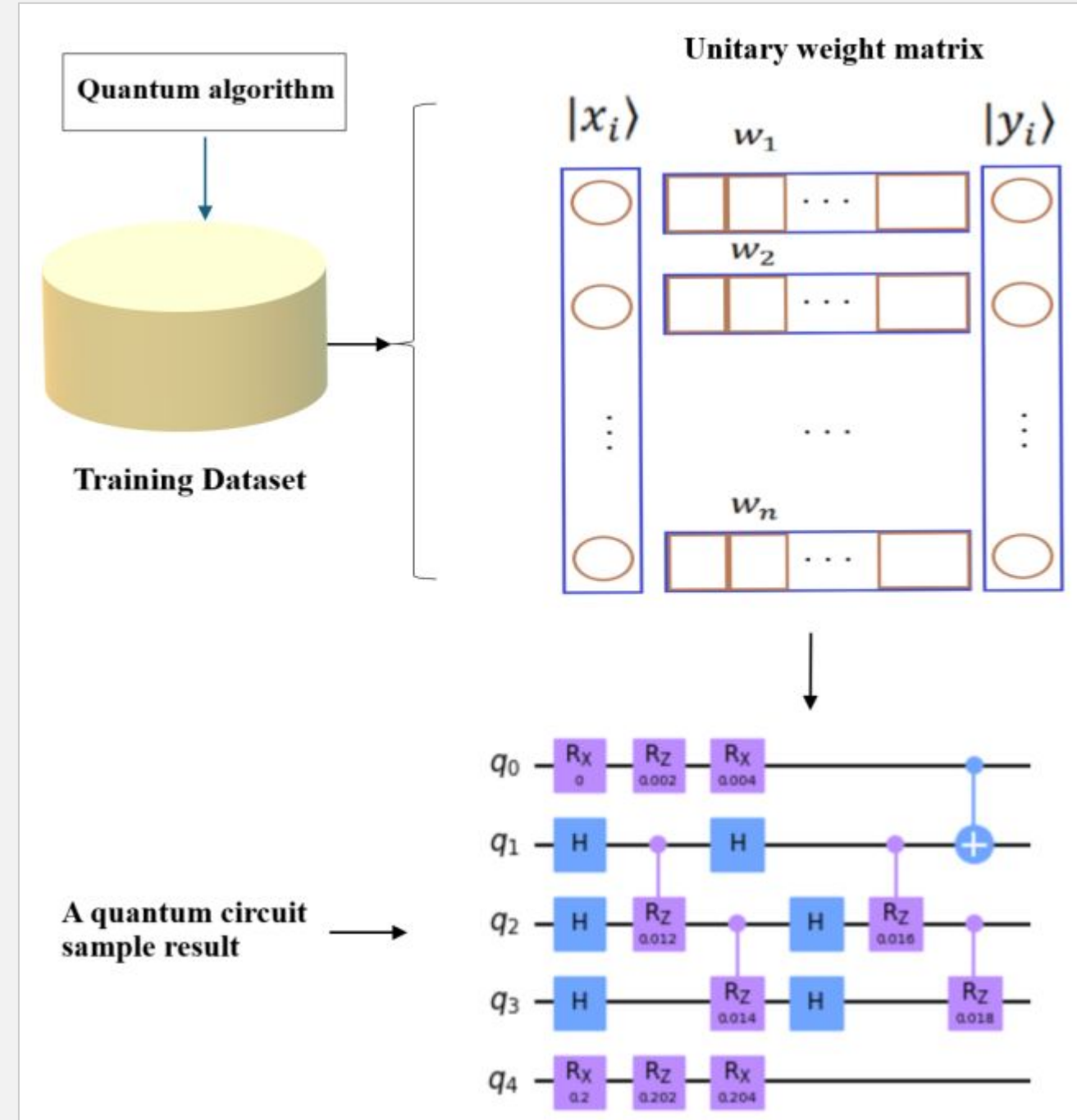


Fig 1: General protocol for quantum circuit synthesis including neural network training and the resulting quantum circuit after the transpilation in Qiskit. [2]

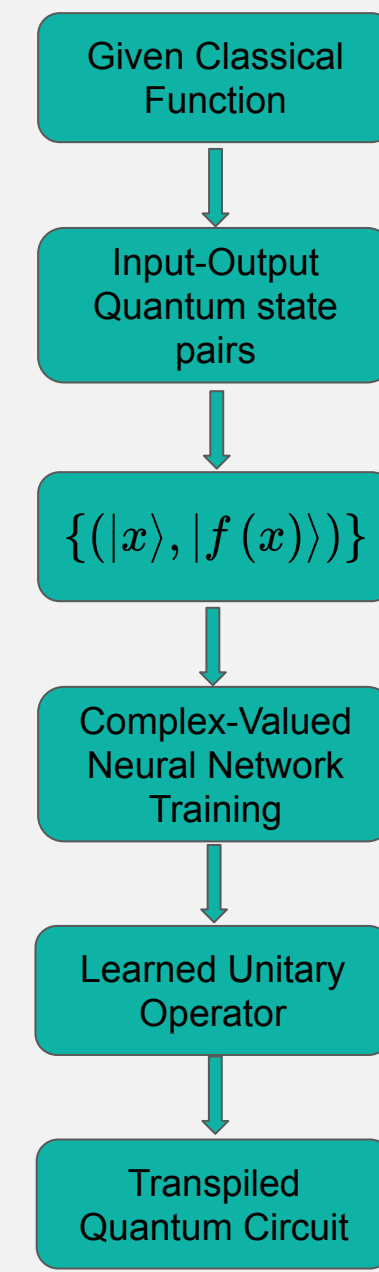


Fig 2: Flow-chart for creating quantum circuit to implement a classical function for a quantum algorithm

2. Prior Work

Consider training on a n qubit circuit, so total number of basis states is, $N = 2^n$. The methodology is as follows [2]:

- Single-layer feed-forward neural network, input state: $|\psi\rangle \in \mathbb{C}^N$, target states: $|\psi_{\text{target}}\rangle \in \mathbb{C}^N$
- Weight matrix ($W \in \mathbb{C}^{N \times N}$) represents the unitary transformation.
- Linear activation and no bias. Forward pass: $|\psi'\rangle = W|\psi\rangle$
- Loss function, $\mathcal{L} : \mathbb{C}^{N \times N} \rightarrow \mathbb{R}$, $\mathcal{L} = ||\psi_{\text{target}}\rangle - |\psi'\rangle||^2$
- **Unitarity enforcement:** Periodic Gram-Schmidt orthogonalization [2]:

$$W^{(k+1)} = \text{GramSchmidt} \left(W^{(k)} - \lambda \nabla_W \mathcal{L} \right)$$

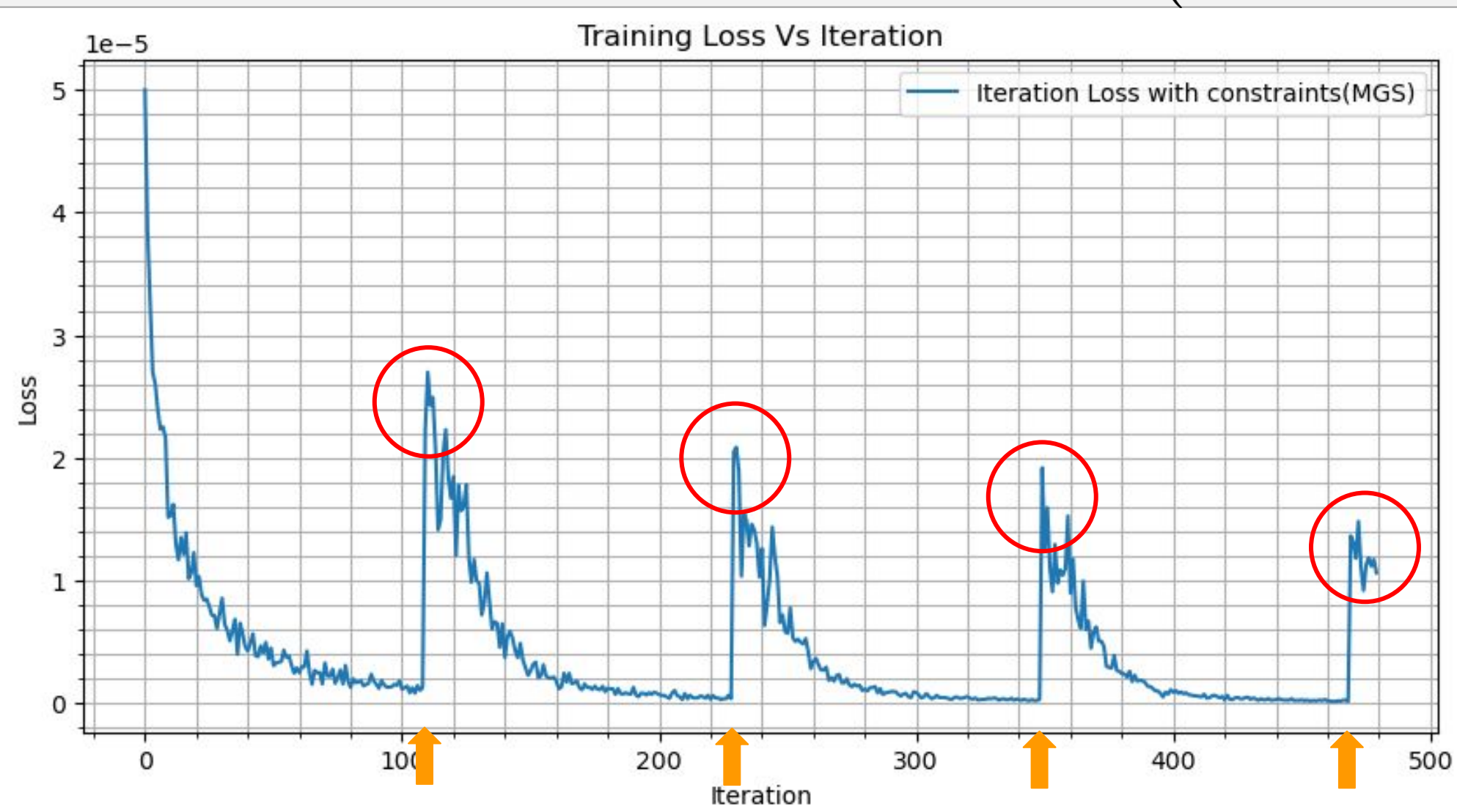


Fig 3: Loss vs Iteration curve of training on a 2-qubit entanglement quantum circuit when Gram-Schmidt orthogonalization is applied to the weight matrix at a fixed interval of epochs to enforce unitarity

Limitations of this approach:

- **Intermittent projection** : loss spikes, unstable training.
- **Numerical instability** and off-manifold weight updates.
- **No smooth optimization** : disrupts gradient-based learning.

3. Methodology: Optimization in Stiefel Manifold

The optimization problem is:

$$\min_W \mathcal{L}(W) \text{ subject to } W^\dagger W = I$$

We constrain weights to the complex Stiefel manifold [3]:

$$M(\mathbb{C}^N) = \{W \in \mathbb{C}^{N \times N} | W^\dagger W = I\}$$

Algorithm:

- Compute Euclidean gradient: $G = \nabla_W \mathcal{L}(W^{(k)})$ using Wirtinger Calculus [5].
- Compute the skew-Hermitian operator:

$$A = G(W^{(k)})^\dagger - W^{(k)} G^\dagger$$

- Update via Cayley transform [4] (smooth & unitary-preserving):

$$W^{(k+1)} = \left(I + \frac{\lambda}{2} A \right)^{-1} \left(I - \frac{\lambda}{2} A \right) W^{(k)}$$

Where $\lambda \in \mathbb{R}$ is the learning rate.

Theoretical Guarantees:

- **Unitarity Preservation:** $W^{(k+1)} \in M(\mathbb{C}^N)$ for all k
- **Descent Property:** $\mathcal{L}(W^{(k+1)}) \leq \mathcal{L}(W^{(k)})$ for all $\lambda > 0$
- **Smooth Optimization:** No jumps or discontinuities.

4. Results

Trained on random qiskit circuits of dimension: {2, 3, 4, 5, 7, 10} qubits.

Metric	Gram-Schmidt	Stiefel Manifold
Unitarity Preservation	Intermittent	Continuous
Training Stability	Discontinuous	Smooth
Final Unitary Error	$\sim 10^{-5}$	$\sim 10^{-12}$
Convergence Speed	Slower	Faster
Numerical Stability	Unstable	Robust

Here we show example of training on a 7-qubit circuit:

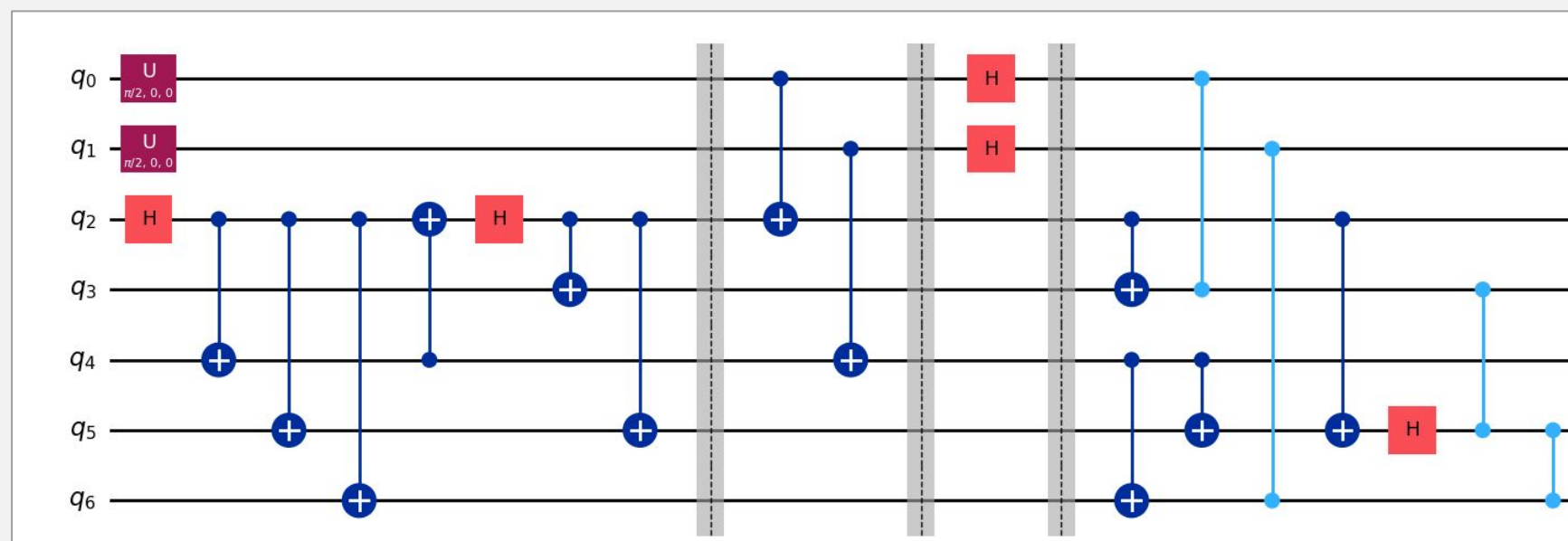


Fig 4: A 7-qubit random quantum circuit from where we collect the input-output quantum state pairs. This is a representation of a quantum algorithm.

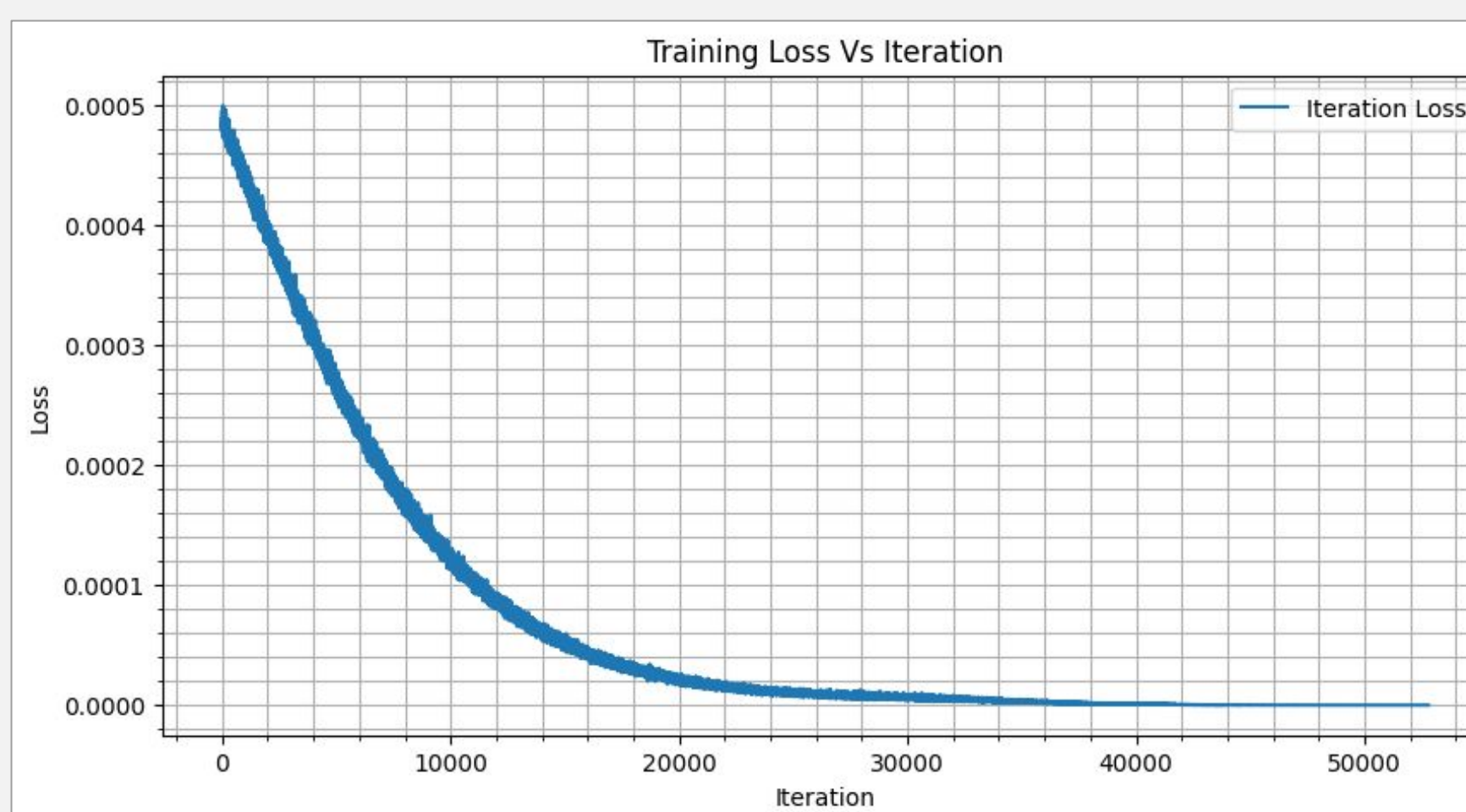


Fig 5: Training loss vs Iteration

- Number of datapoints used: 15000, number of epochs = 150, learning rate = 0.8
- No hidden layer, no bias, linear activation
- 128 neurons in both input and output layer.
- We observe a smoothly decreasing loss curve.

To analyse the training performance we use two metrics - Fidelity and Unitary Error.

Fidelity:

- Fidelity measures how close two quantum states or operations are.
- Ranges from 0 (completely different) to 1 (identical). We want the fidelity to be 1 at the end of training.
- We calculate fidelity between learned unitary matrix W_{final} and target unitary U_{target} as follows:

$$F = \frac{1}{2^n} \left| \text{Tr} \left(W_{\text{final}}^\dagger U_{\text{target}} \right) \right|$$

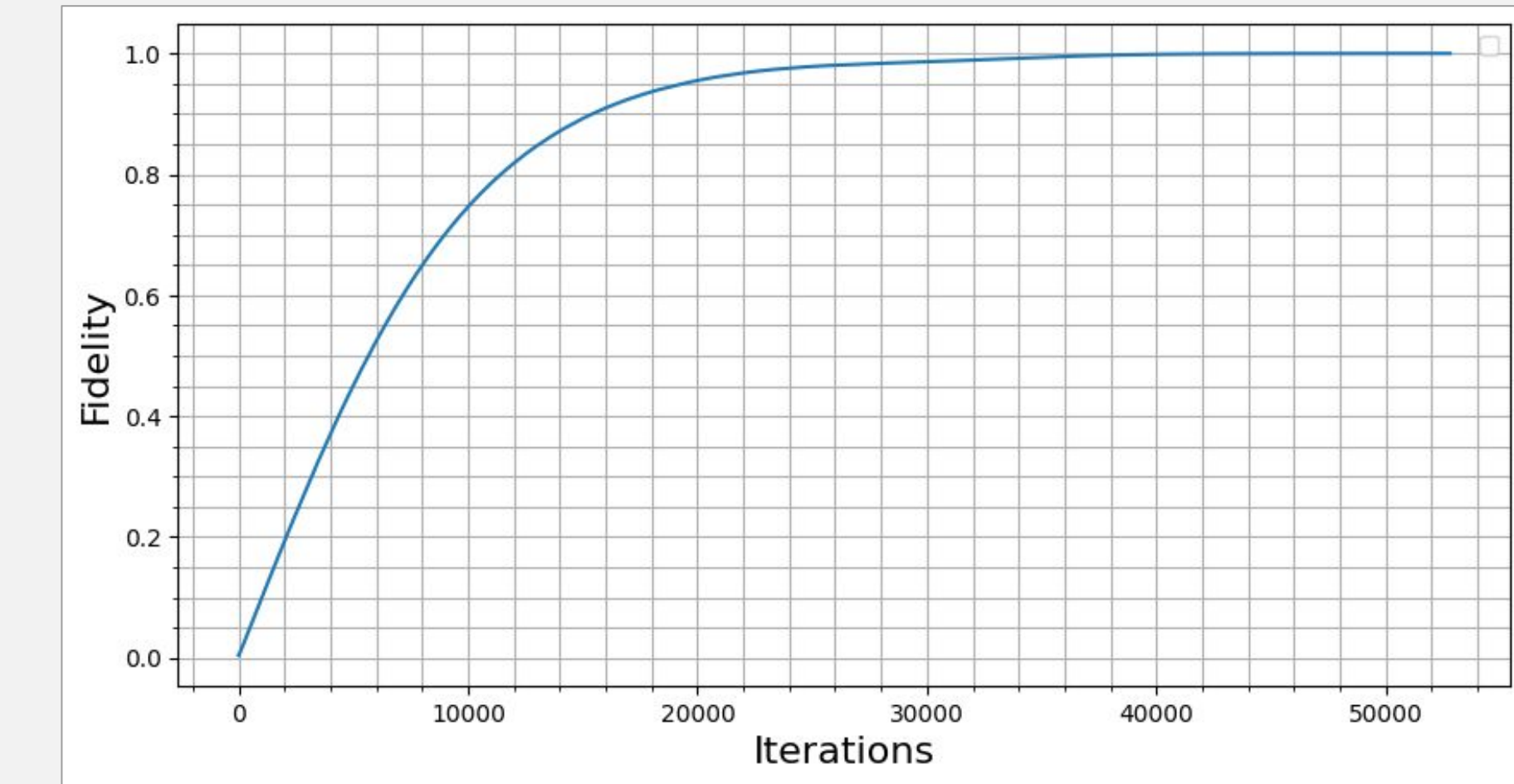


Fig 6: Fidelity vs Iteration

Here, we can observe that the fidelity is reaching 1 as the training progresses (desired).

Unitary Error:

It is calculated at each update to make sure that the weight matrix remains in the Stiefel Manifold after each iteration. We observe that the unitary error is in order of 10^{-11}

$$||W^\dagger W - I||_F^2 \sim 10^{-11}$$

5. Impact of Multiple Hidden Layers

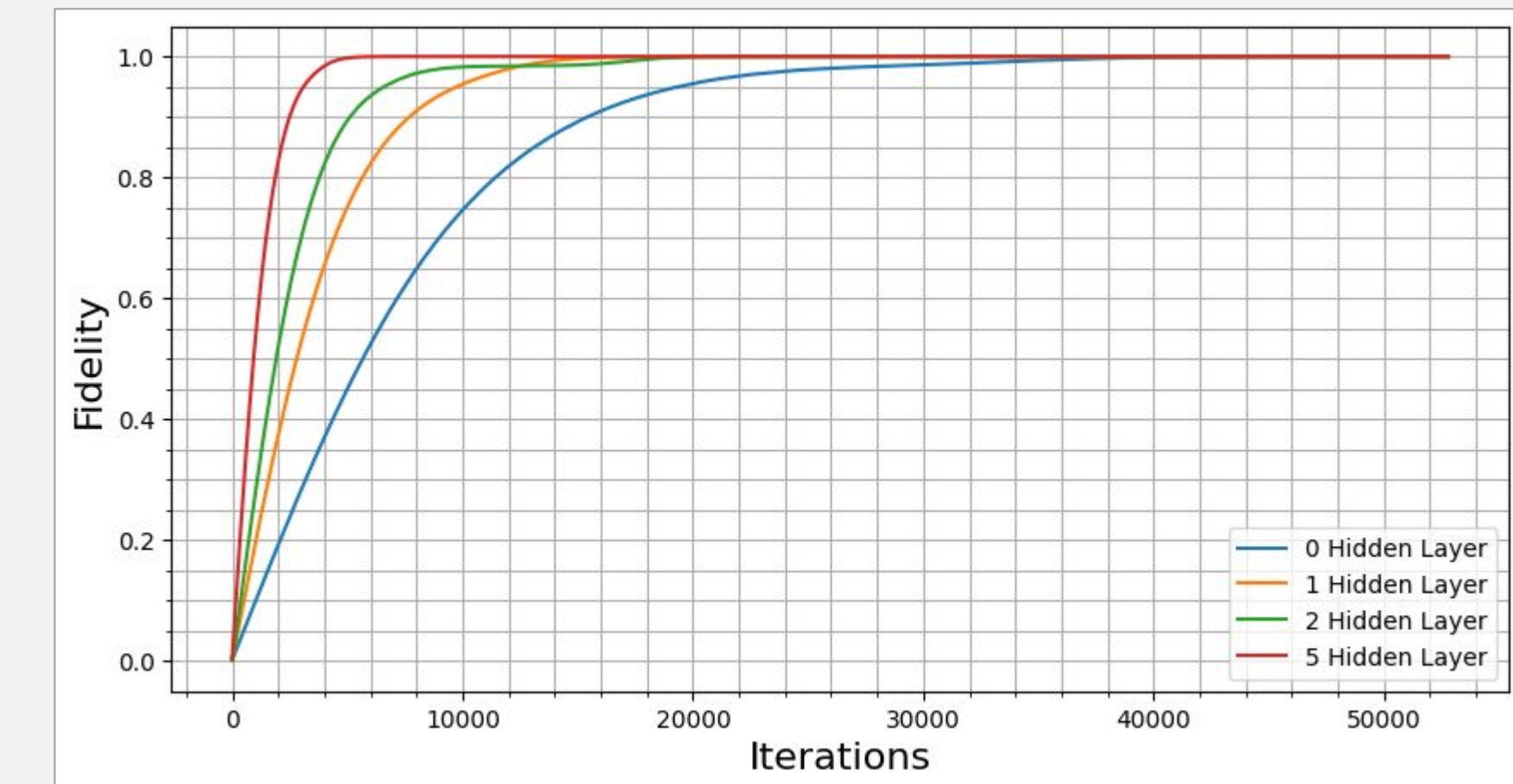


Fig 7: 7-qubit circuit training fidelity for: 0HL, 1HL, 2HL, 5HL (HL=Hidden Layer)

- More hidden layers $\rightarrow$ faster convergence.
- Better representation capacity & hierarchical learning.
- Overall unitary: $W_{\text{net}} = \Pi W_i$ stays unitary

5. Future Work

- Optimal hidden layer count vs. saturation.
- Minimum training data requirements for unit fidelity.
- Scaling to 100+ qubit systems.
- Trade-off between depth and training time.

6. References

- [1] M. A. Nielsen & I. L. Chuang, Quantum Computation and Quantum Information. Cambridge Univ. Press.
- [2] M. Zomorodi et al., "Optimal Quantum Circuit Design via Unitary Neural Networks," arXiv:2408.13211 (2024).
- [3] P.-A. Absil, R. Mahony, & R. Sepulchre, Optimization Algorithms on Matrix Manifolds, Princeton Univ. Press (2009).
- [4] Wisdom et al., Full-Capacity Unitary RNNs, NeurIPS 2016.
- [5] Kreutz-Delgado, K. The Complex Gradient Operator and the CR-Calculus. arXiv:0906.4835, 2009.